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DIGITAL LOGIC

Assignment of Chapter Two (C)

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1). Fill-in-the-blank Questions

1 The simplification of the logic algebra contains three common methods, which are algebra simplification, Karnaugh map simplification, and Quine-McCluskey tabular simplification.

2 Each square in a Karnaugh map represents a Minterm.

3 A Karnaugh map with n variables consists of 2^n (how many) squares.

4 “The Karnaugh map is suitable to simplify the functions with many variables.” This statement is false (true/false).

B). Use the algebra simplification method to find the simplest “AND-OR” expressions of the following logic functions.

3

$$\begin{aligned} \square 1) \quad F &= \bar{A} \bar{B} + \bar{B} \bar{C} + BC + AB \\ &= \bar{A} \bar{B} (C + \bar{C}) + (A + \bar{A}) BC + \bar{B} \bar{C} + AB \\ &= \bar{A} \bar{B} C + \bar{A} \bar{B} \bar{C} + ABC + \bar{A} BC + \bar{B} \bar{C} + AB \\ &= \bar{A} \bar{B} C + \bar{B} \bar{C} + AB + \bar{A} BC \\ &= \bar{A} C (\bar{B} + B) + \bar{B} \bar{C} + AB \\ &= \bar{A} C + \bar{B} \bar{C} + AB \end{aligned}$$

Method 2

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□ 1)
$$\begin{aligned} F &= \bar{A} \bar{B} + \bar{B} \bar{C} + BC + AB \\ &= \bar{A} \bar{B} + (A + \bar{A}) \bar{B} \bar{C} + BC + AB (C + \bar{C}) \\ &= \bar{A} \bar{B} + A \bar{B} \bar{C} + \bar{A} \bar{B} \bar{C} + BC + AB C + AB \bar{C} \\ &= \bar{A} \bar{B} + A \bar{B} \bar{C} + BC + AB \bar{C} \\ &= \bar{A} \bar{B} + A \bar{C} + BC \end{aligned}$$

2)

5

$$\square F = (A + B + C)(\bar{A} + B)(A + B + \bar{C})(C + D)$$

$$F' = ABC + \bar{A}B + AB\bar{C} + CD$$

$$= AB(C + \bar{C}) + \bar{A}B + CD$$

$$= AB + \bar{A}B + CD$$

$$= B + CD$$

$$F = B(C+D) = BC+BD$$

2) Method 2

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$$\begin{aligned}\square F &= (A + B + C)(\bar{A} + B)(A + B + \bar{C})(C + D) \\ &= (\bar{A} + B)(A + B)(C + D) \\ &= B(C + D) \\ &= BC + BD\end{aligned}$$

3)

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$$\begin{aligned}\square F &= AB\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{C} \\ &= AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{C} \\ &= A\bar{C}(B + \bar{B}) + \bar{A}(BC + \bar{C}) \\ &= A\bar{C} + \bar{A}(B + \bar{C}) \\ &= A\bar{C} + \bar{A}B + \bar{A}\bar{C} \\ &= \bar{A}B + \bar{C}\end{aligned}$$

3) Method 2

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$$\begin{aligned}\square F &= AB\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{C} \\ &= AB\bar{C} + \bar{A}BC + (A\bar{B} + \bar{A})\bar{C} \\ &= AB\bar{C} + \bar{A}BC + (\bar{B} + \bar{A})\bar{C} \\ &= AB\bar{C} + \bar{A}BC + \overline{AB}\bar{C} \\ &= \bar{C} + \bar{A}BC = \bar{C} + \bar{A}B\end{aligned}$$

□ 4 $F = AB + \overline{\overline{A}BC} + BC$

A:

$$\begin{aligned}
 F &= AB + \overline{\overline{A}BC} + BC = AB + (\overline{\overline{A}B} + B)C \\
 &= AB + \overline{(\overline{A} + B)}C = AB + \overline{A}C + BC \\
 &= AB + \overline{A}C
 \end{aligned}$$

□ 5. $F = A\overline{B} + B + BCD$

$$\begin{aligned} (2) \quad F &= A\overline{B} + B + BCD = A\overline{B} + B(1 + CD) \\ &= A\overline{B} + B = A + B \end{aligned}$$

□ 6 $F = (A + B + C)(\bar{A} + B)(A + B + \bar{C})$

A:

$$\begin{aligned} (3) \quad F &= (A + B + C)(\bar{A} + B)(A + B + \bar{C}) \\ &= (A + B)(\bar{A} + B) \\ &= B \end{aligned}$$

Use the Karnaugh Map simplification method to find the simplest “AND-OR” and “OR-AND” expressions of the following logic functions.

$$F(A,B,C,D) = \overline{A}\overline{B} + \overline{A}\overline{C}D + AC + B\overline{C}$$

A:

$$(1) \quad F(A,B,C,D) = \overline{A}\overline{B} + \overline{A}\overline{C}D + AC + B\overline{C}$$

AB \ CD		00	01	11	10
CD	00	1	1	1	
	01	1	1	1	
	11	1		1	1
	10	1		1	1

Therefore

$$\begin{aligned}
 F(A,B,C,D) &= \overline{A}\overline{B} + \overline{A}\overline{C}D + AC + B\overline{C} \\
 &= \overline{A}\overline{B} + AC + B\overline{C} \\
 &= (A + \overline{B} + \overline{C})(\overline{A} + B + C)
 \end{aligned}$$

$$(2) \quad F(A,B,C,D) = \prod M(2,4,6,10,11,12,13,14,15)$$

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$$F(A,B,C,D) = \prod M(2,4,6,10,11,12,13,14,15) = \sum m(0,1,3,5,7,8,9)$$

AB \ CD	00	01	11	10
00	1			1
01	1	1		1
11	1	1		
10				

$$F(A,B,C,D) = \bar{A}D + \bar{B}\bar{C}$$

$$\bar{F}(A,B,C,D) = AB + AC + B\bar{D} + C\bar{D}$$

$$F(A,B,C,D) = (\bar{A} + \bar{B})(\bar{A} + \bar{C})(\bar{B} + D)(\bar{C} + D)$$